

Shock

What is meant by shock? Let's return to our two intensive care unit (ICU) patients.

Patient A: BP 100/40, HR 114, UOP < 10 mL

Patient B: BP 80/40, HR 85, UOP 50 mL/hr

Which patient is in shock? Patient A? Patient B? Both?

WHAT IS SHOCK?

Often the bias is to define shock simply as a low blood pressure. While this is certainly part of the equation, it is not the whole picture.

Why is low blood pressure usually the first thing we think about when we think about shock? Usually, it is because this is the first parameter that presents itself – the monitor starts beeping, alarms are going off, and people around us start getting concerned. However, we have to remember that a low blood pressure, specifically a low mean arterial pressure (MAP), may or may not mean that a patient is in shock ([Fig. 2.1](#)).

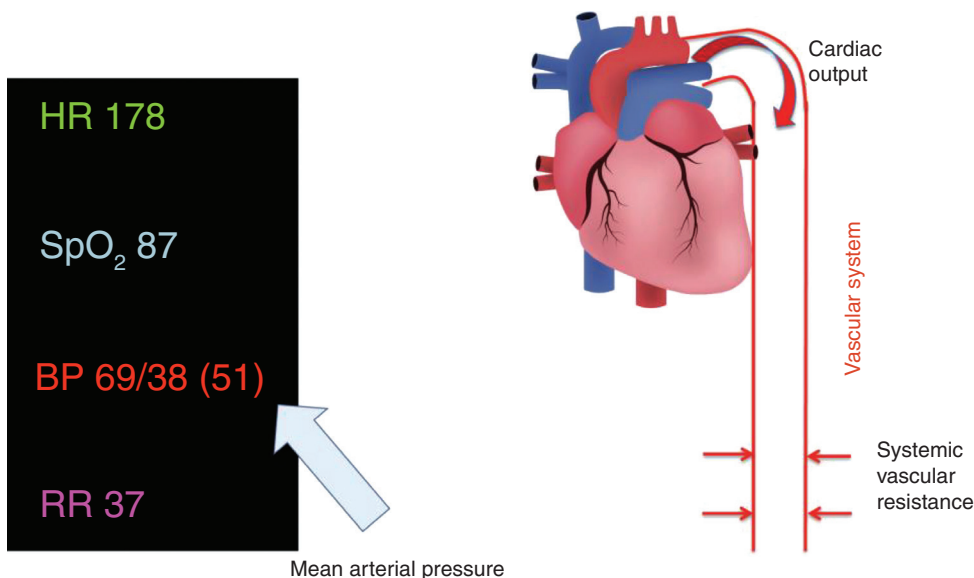


FIG. 2.1 Shock: blood pressure versus cardiac output. (Modified from [Ody_Stocker/Shutterstock.com](#).)

WHAT IS THE DIFFERENCE BETWEEN LOW BLOOD PRESSURE AND SHOCK?

Shock is defined as circulatory failure causing inadequate delivery of oxygen to meet oxygen consumption requirements, leading to cellular and tissue hypoxia. It all comes back to $\text{DO}_2:\text{VO}_2$. When DO_2 is not adequate to keep up with VO_2 , shock ensues.

This is an important distinction from blood pressure. A low MAP can be evidence of a low cardiac output and, thus, a deficiency of $\text{DO}_2:\text{VO}_2$, but it does not mean that the patient is in shock.

Rather, the MAP is an observed value calculated by the systolic and diastolic blood pressure, which is ultimately a factor of cardiac output, central venous pressure (CVP) to a lesser degree, and systemic vascular resistance (SVR) (Fig. 2.2).

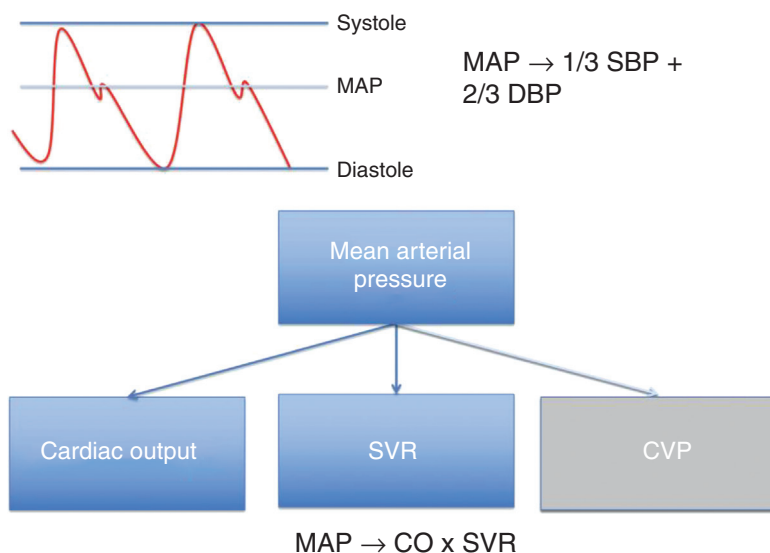


FIG. 2.2 Mean arterial pressure as a surrogate for cardiac output. CVP, Central venous pressure; SVR, systemic vascular resistance.

We are making this distinction because, ultimately, it does not matter to end organs what the MAP is, but rather how much oxygen is being delivered. It is tempting to chase after specific MAP targets, but these targets only matter to the degree that they are an observed manifestation of cardiac output and ultimately DO_2 .

If you measured the blood pressure of 100 random people, you would notice that there is a vast difference in the overall MAP. Some would have a very high MAP and some may have a very low MAP. The people with the lower MAP likely are not in shock. They may have completely adequate perfusion, normal lactate levels, great urine output, and overall, adequate DO_2 .

WHAT IS THE RELATIONSHIP OF MAP TO CARDIAC OUTPUT?

There are certainly some people who if you measured their MAP throughout the course of their normal day (especially during sleep), you might notice a MAP of 65, 60, 55 mmHg, or even below. What is going on with these people? Are we catching shock that would normally go unrecognized?

Clearly not. What we are uncovering here is that MAP only relates to low oxygen delivery to the degree that it is an observable indicator of cardiac output. The reason why some people can maintain adequate oxygen delivery and cardiac output at a lower MAP is SVR.

SVR represents the overall tone of the blood vessels. The normal relation of MAP to SVR is traditionally expressed as:

$$\text{MAP} \rightarrow \text{CO} \times \text{SVR}$$

Thus, we can imagine that the same cardiac output, as represented by the green line, could correspond to very different MAPs, based on what the overall SVR is, as illustrated in Fig. 2.3.

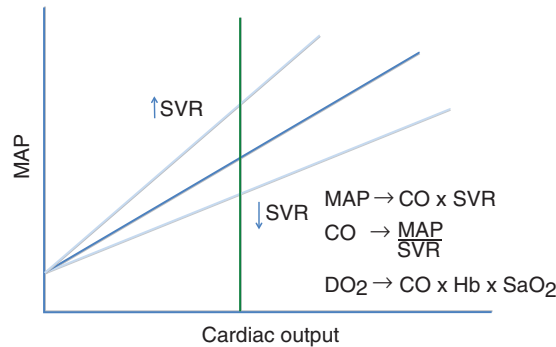


FIG. 2.3 The relationship of mean arterial pressure and cardiac output: the role of systemic vascular resistance. *MAP*, Mean arterial pressure; *SVR*, systemic vascular resistance.

Rather than stating that MAP is a function of CO and SVR, we can consider reframing the relationship as:

$$\text{CO} \rightarrow \frac{\text{MAP}}{\text{SVR}}$$

In this regard, we can see that MAP is the observable manifestation of cardiac output as a function of SVR maintained by the blood vessels. SVR is a derived value and MAP is a measured value. Ultimately, both help to describe the cardiac output and, ultimately, the DO_2 .

REFRAMING HOW WE DEFINE SHOCK

Let's return to our $\text{DO}_2:\text{VO}_2$ relationship. Remember that as $\text{DO}_2:\text{VO}_2$ decreases to below 2:1, we begin to head toward cardiopulmonary and total system collapse (Fig. 2.4).

In order to determine how oxygen is being delivered, we must recognize inadequate delivery of oxygen above all else. If a patient has a normal blood pressure because the SVR is high but oxygen is not being delivered, he or she may be in shock. If a different patient has lower blood pressure because of a lower SVR, but oxygen is being delivered, urine output is adequate, mental status is maintained, and all tissues have evidence of adequate perfusion, shock may not be present, regardless of what the blood pressure is.

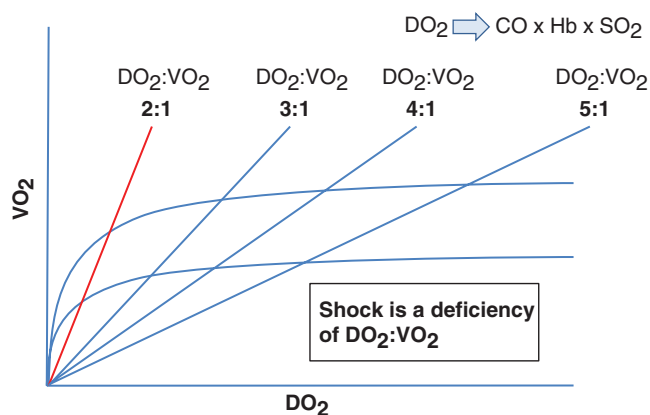


FIG. 2.4 Shock and $DO_2:VO_2$

HOW DO WE CLASSIFY SHOCK?

Now that we have the framework to recognize shock as a deficiency of oxygen delivery, let's turn our attention to identifying the cause of shock. You will come to appreciate this as the essential piece to managing a patient in shock. Once you know what you are up against, you will be in a much better position to intervene and improve the outcome for your patient.

The traditional classification of shock is based on the relative contributions of low cardiac output and SVR: cardiogenic, obstructive, hypovolemic, and distributive (Fig. 2.5).

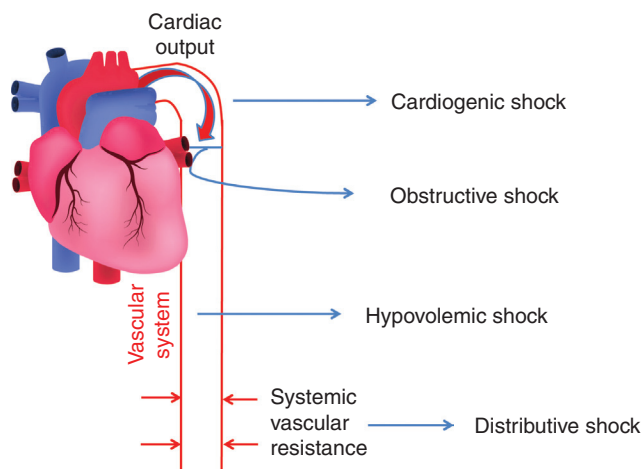


FIG. 2.5 Classification of shock. (Modified from [Ody_Stocker/Shutterstock.com](#).)

It is often tempting to think of these as fixed classifications, with each patient being assigned a designation when they hit the ICU just as they are assigned a medical record number. Your patient with NSTEMI? Cardiogenic shock. Your patient with urosepsis? Distributive shock.

Unfortunately, it is never this simple – each classification can have a multitude of mechanisms, pathophysiologic processes, and etiologies associated. Pericardial tamponade, pulmonary embolism, and pneumothorax can all cause obstructive shock but obstruct in very different ways. The distributive shock caused by anaphylaxis is very different from distributive shock due to sedation effect or neurogenic shock. Even distributive shock due to sepsis can be very different both clinically and on the cellular level when caused by different pathogens (Fig. 2.6).

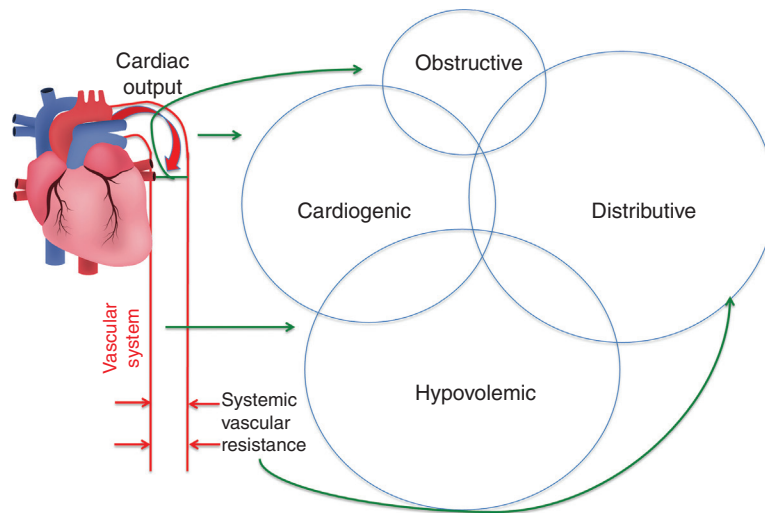


FIG. 2.6 Shock as a clinical spectrum. (Modified from [Ody_Stocker/Shutterstock.com.](#))

These etiologies can often run together, where components of multiple types of shock can be exhibited for a single patient at different points in the clinical course.

Ultimately, classifications are only as useful as their ability to help understand the physiology specific to the patient.

UNCOVERING THE TRUE ETIOLOGY OF SHOCK

The example of distributive shock can be telling. Often when distributive shock is diagnosed, we correctly identify a dysregulation of SVR. However, it is often tempting to think of vascular resistance as the homogenous capability of the blood vessels to provide vascular tone.

However, this lack of tone may be complex in its etiology. Vascular tone is maintained by a vascular endothelium and vascular smooth muscle, with a variety of hormones, neurotransmitters, and vasoactive factors maintaining the endothelium and a host of receptors and cofactors affecting the vascular smooth muscle ([Fig. 2.7](#)).

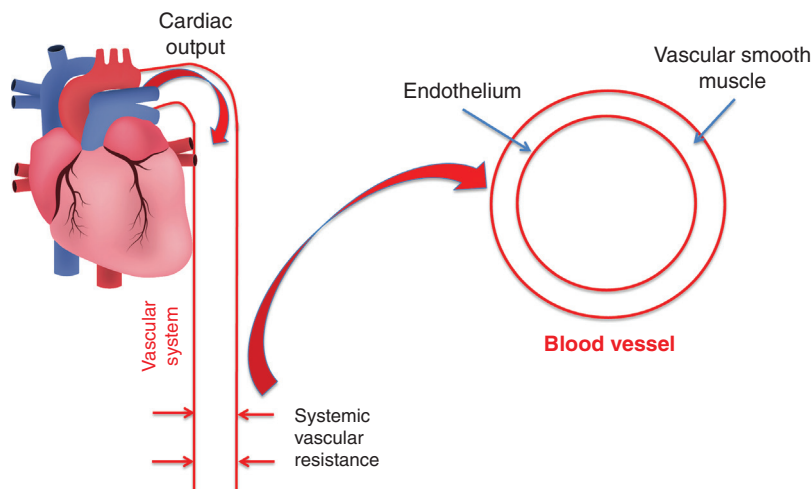


FIG. 2.7 The components of vascular tone. (Modified from [Ody_Stocker/Shutterstock.com.](#))

This complex system allows for patients to differ not only in their presentation, but their response to therapy. A patient whose loss of vascular tone is primarily a function of decreased stimulation by the adrenergic nervous system may respond particularly well to epinephrine, while a patient whose tone is primarily due to a relative deficiency in antidiuretic hormone may respond much better to vasopressin.

In reality there are a host of deficiencies that can be responsible for this loss of vascular tone, which are illustrated in Fig. 2.8.

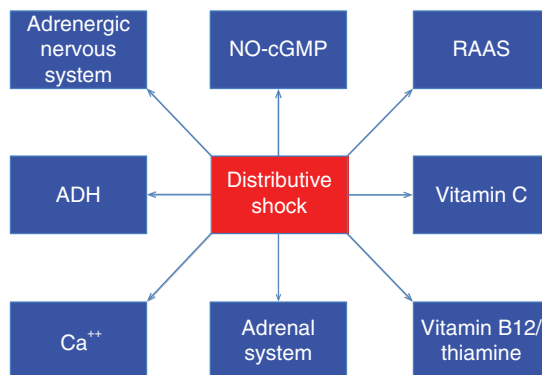


FIG. 2.8 The multiple components required to maintain vascular tone. *ADH*, Antidiuretic hormone; *NO-cGMP*, nitric oxide cyclic guanosine monophosphate; *RAAS*, renin-angiotensin-aldosterone system.

We often note the effect clinically – that our patient “responded really well to vasopressin” or “really liked the calcium.” Overall, this serves to remind us that shock can be complex in its etiology, and that the classifications that we set up are designed to better understand the physiology that is unique to the patient.

OPTIMIZING CARDIAC OUTPUT IN SHOCK

So what do we do about shock? How can we improve our cardiac output? The traditional consideration is to put it into the context of preload, afterload, and contractility (Fig. 2.9).

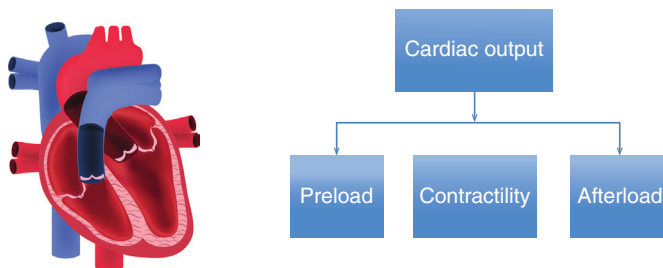


FIG. 2.9 Cardiac output and the contribution of preload, afterload, and contractility. (Modified from [Ody_Stocker/Shutterstock.com](https://www.shutterstock.com/user/Ody_Stocker).)

Let’s discuss the individual components.

Preload and cardiac output

When considering preload, consider a slingshot. If you were to load a pebble on a slingshot and draw back the band, the pebble would fly a certain distance. Now, let’s envision pulling the band back further ... the pebble will fly further. Draw back further still, and the pebble will fly further.

However, you can also envision that as we continue to draw back further and further, the distance the pebble will fly begins to become less and less. After a certain point, we experience progressively diminishing returns for further increases. In fact, if we were to plot this relationship on a graph, you might expect something as follows, where there is higher responsiveness to preload initially, followed by less responsiveness with progressive increases in preload (Fig. 2.10).

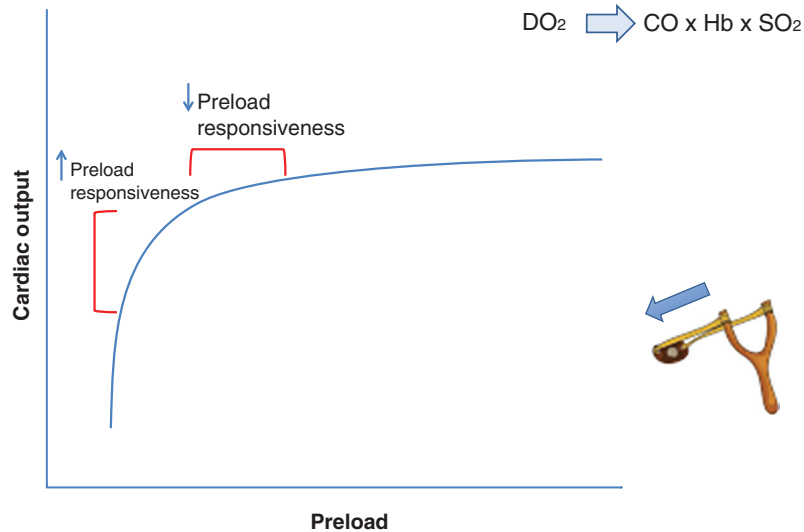


FIG. 2.10 Cardiac output as a function of preload. (Modified from Pashigorov/PREMIUM Stock Vector/Shutterstock.)

This relationship is what we observe with regard to preload and cardiac output. There is a point where preload corresponds to a significant increase in cardiac output, and a point where there is less responsiveness of cardiac output to preload.

What is the effect of afterload?

Conversely, let's consider the effect of afterload. In this case, envision the effect of pushing a box across the room. Now, imagine adding weight to the box to make it progressively heavier. Initially, the heavier box barely impacts your ability to push the box across the room. However, at a certain point, there will be a rapid decline in the ability to push this box.

Just like different people are able to push different size boxes, the response to afterload can be varied based on the capabilities of the heart. Regardless of the overall tolerance to afterload, the pattern is a period of minimal effects of afterload, followed by a sharp decline in cardiac output (Fig. 2.11).

Preload and afterload in the setting of the right and left heart

Although this convention of preload and afterload is often applied to the heart as a whole, it can sometimes be helpful to think of the effect of preload and afterload as they relate to the left and right heart.

Specifically, the right heart offers the opportunity to consider the effect of preload and afterload in a unique way, which may help to explain the clinical characteristics that we observe at the bedside (Fig. 2.12).

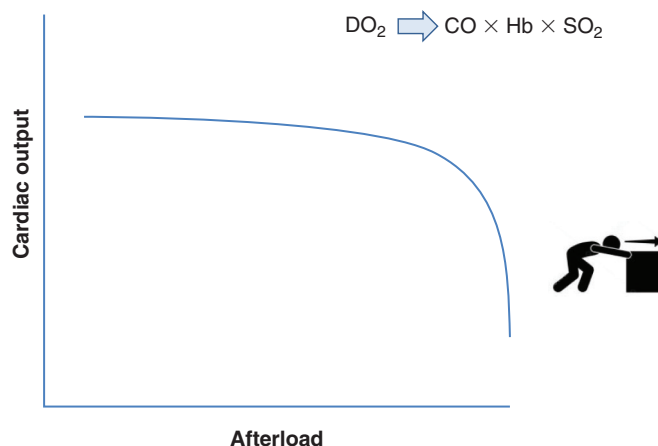


FIG. 2.11 Cardiac output as a function of afterload. (Modified from Leremy/Stock Illustrations/Shutterstock.)

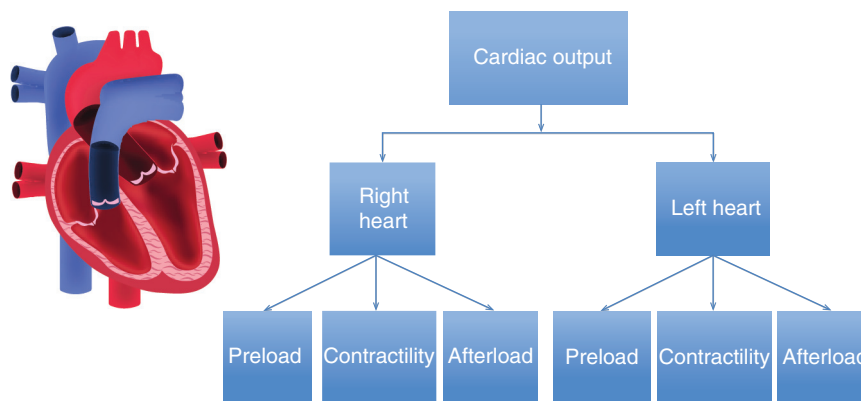


FIG. 2.12 Preload, afterload, and contractility of the left and right heart. (Modified from Ody_Stocker/Shutterstock.com.)

The right heart behaves very differently from the left heart. Where the left heart is designed to push blood through the entire systemic circulation, the right heart is primarily designed to pump blood through the pulmonary circulation. This difference in function is reflected in differences in form – in general, it is less muscular, contracts on a different axis, and exhibits different responses to filling pressures.

Being less muscular, the right heart exhibits a less robust response to preload with an abrupt stop due to limitations of the pericardium. The effect of preload on the left and right heart is illustrated in Fig. 2.13.

This means that there is a limit to how much volume can be administered to augment the right heart output. At a certain point, the stretch afforded by additional volume only translates into more dilation of the right ventricular cavity, which can have an adverse effect on cardiac output.

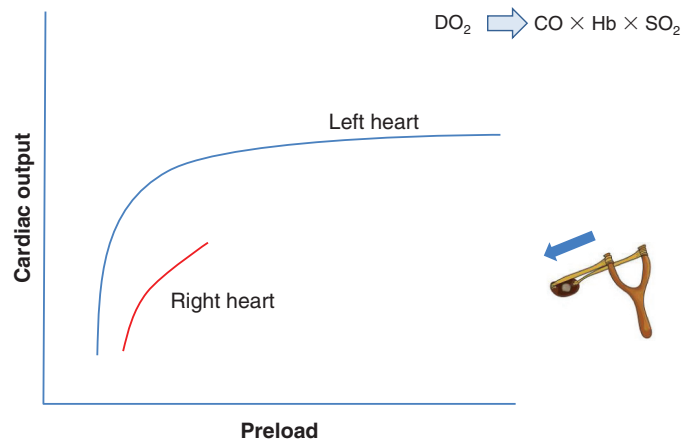


FIG. 2.13 Preload effect for the left and right heart. (Modified from Pashigorov/PREMIUM Stock Vector/Shutterstock.)

How does excess preload adversely affect cardiac output in a failing right ventricle?

When preload is given to a right ventricle that is no longer preload responsive, the right ventricle can only expand across the septum, which in turn compresses the left ventricle (Fig. 2.14).

Doing so may actually drop the left ventricular preload, which can be observed clinically – when fluid is administered in the setting of right ventricular failure, the blood pressure may actually go down.

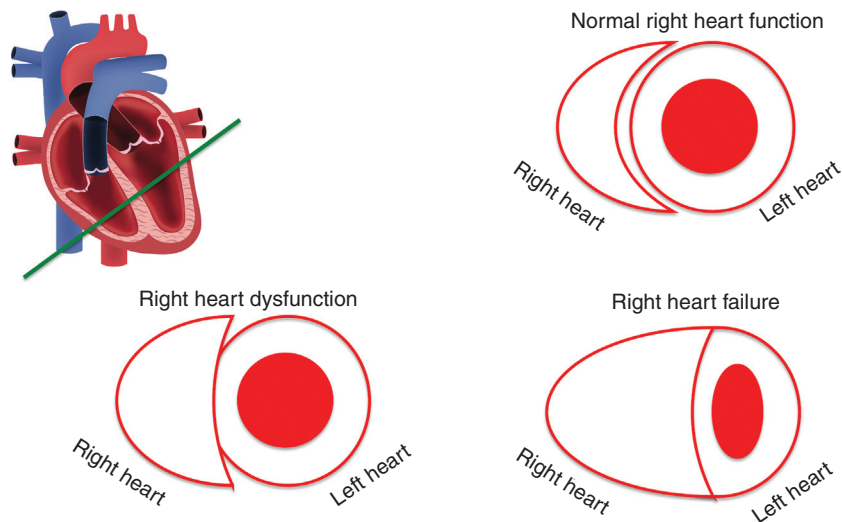


FIG. 2.14 Effect of preload on failing right ventricle. (Modified from Ody_Stocker/Shutterstock.com.)

Afterload and the right ventricle

In a similar manner to the unique effects of preload on the right ventricle, afterload can have a profound impact on the overall function of the right ventricle. Being less muscular and contracting on a different axis, the right ventricle is exquisitely sensitive to the effects of afterload, with a sharp drop-off in function with even modest increases in afterload (Fig. 2.15).

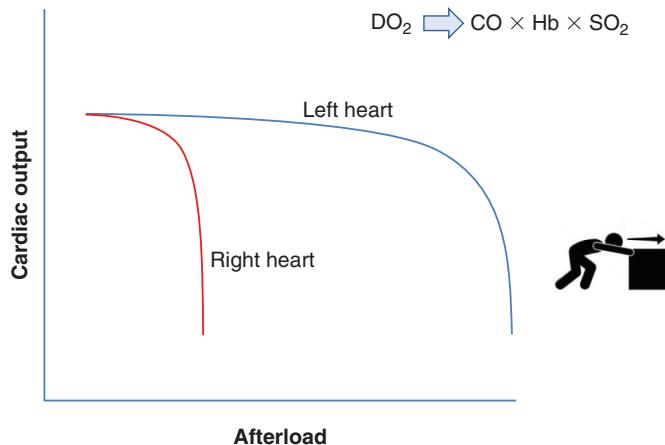


FIG. 2.15 Cardiac output as a function of afterload for the left and right heart. (Modified from Leremy/Stock Illustrations /Shutterstock.)

Variables increasing right ventricular afterload

Since the right ventricle is pumping through the pulmonary circulation, anything that increases pulmonary vascular resistance (PVR) increases this afterload and carries with it the risk of worsening right ventricular output.

Can this happen for conditions that are commonly encountered in the ICU? Absolutely! Let's review some of the factors that increase PVR, which certainly are recognizable for many patients you may encounter in the ICU:

1. Hypercapnia (elevated CO_2)
2. Hypoxia
3. Acidosis
4. Hypothermia
5. Decreased lung compliance (from pneumonia, edema, atelectasis, etc.)
6. Elevated adrenergic tone

You can quickly envision the many situations that you might encounter in the ICU where right ventricular afterload can escalate and you can observe a rapid decline in right heart function.

HOW CAN WE USE THIS MODEL OF SHOCK TO IMPROVE THE MANAGEMENT OF OUR PATIENTS?

Models are only helpful in the way they help us to approach and think about what we are observing clinically.

Let's say you see a patient in the ICU with a low blood pressure. What is our approach to this patient?

First, let's evaluate the patient. Is this blood pressure allowing for adequate delivery of oxygen? Are they making adequate urine? Are they awake and talking? Are they warm and well perfused? Are there lab abnormalities?

If they are in shock, the question becomes what is the shock caused by? Are their clinical and historical data pointing to cardiogenic? Distributive? Obstructive? Or is this perhaps a mixed picture with components of more than one type?

Now we are faced with what do we do about shock. Let's address the role of two of the most common interventions in the ICU for shock – intravenous fluid and vasopressor administration.

Rationale for fluid administration in shock

The entire aim of the administration of fluid in shock is to increase preload to the left ventricle and thus increase cardiac output. But let's think through what this scenario looks like and what assumptions must be true for this to occur.

Even though the intended effect is left ventricular output, we are administering fluid to the venous side of the circulation. The following assumptions must be true for this to translate to an increase in left ventricular output:

1. Enough fluid must stay in the intravascular space to increase right ventricular preload.
2. The preload must increase the right ventricular output.
3. The increase in right ventricular output must be significant enough to augment left ventricular preload.
4. The left ventricle responsive to preload.

Often these assumptions are all true; however, if they are not, then the effect of this additional volume only increases the hydrostatic pressure in the venous system, causing the buildup of fluid in the interstitium and edema (Fig. 2.16).

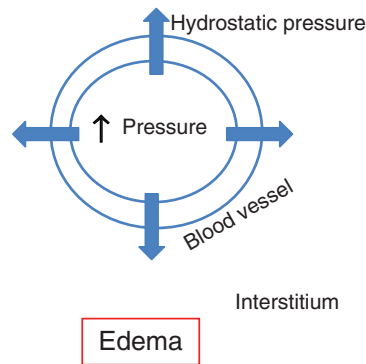


FIG. 2.16 Increased hydrostatic pressure and formation of tissue edema

What is the clinical effect of this edema? Sometimes it is trivial, but for the majority of tissues and organ systems (brain, lungs, heart gut, liver, kidneys, etc.), edema worsens the overall function.

Does this mean that fluids should not be administered in shock? Certainly not! Rather, the point of this exercise is to highlight that our goal should be to optimize DO_2 . The way fluids are able to accomplish this is through optimization of cardiac output by increasing preload. Administration of fluid for the mitigation of shock should be done in this context, and we should always assess the response to fluids for evidence that an improvement in oxygen delivery has occurred.

How can pressors help in shock?

For the patient in shock who is unresponsive to fluids, the next step is often to consider the use of vasopressors. The reasoning is sound – if there is a vasodilatory component of shock, vasopressors can improve vascular tone, increasing SVR and, therefore, the blood pressure.

Does this help to mitigate shock? Remember that shock is harmful to the degree that it impairs the ability to deliver oxygen sufficient for the body's needs. Increasing the MAP by increasing SVR does not necessarily translate into an improved DO_2 .

Instead, we have to remember that increasing SVR can help to optimize DO_2 by improving vascular tone, and ultimately optimizing venous return and cardiac output. If we are not achieving this, we will see an improvement in MAP but not necessarily an improvement in the delivery of oxygen.

How can we know we are improving cardiac output with our interventions and not just increasing our blood pressure? The answer lies in the physiologic response. For every intervention that we perform, we should ask ourselves “is the intervention that I just performed helping the patient in some meaningful way?” We should ask ourselves if there is an improvement in urine output, a decrease in lactic acid level, or an improvement in liver function tests. Evaluation of the patient before and after an intervention can provide powerful insight into what is going on and can help to guide what interventions are appropriate.

So which pressor should I use?

Hopefully by now, it should be apparent that shock is a complex phenomenon, with a variety of etiologies, manifestations, and physiologic phenomena. It could be classified as primarily cardiogenic, distributive, obstructive, or hypovolemic and could have very different responses to therapy.

To better clarify the effect of pressors in shock, we should understand the effect they have on the cardiac contractility, the systemic vasculature, and importantly on the pulmonary vasculature, which will be critical in the case of any right ventricular failure.

What will follow is not a comprehensive overview but, rather a framework that can be used to approach some of the most common pressors/inotropes utilized when managing a patient in shock.

Norepinephrine

Norepinephrine is one of the most common pressors used. It provides vasoconstriction as well as some inotropy at higher doses, as denoted by the *green arrows*. In this regard, it can be a very useful pressor.

However, note that it can increase the PVR, denoted by the *red arrow*. In patients with a struggling right ventricle, with evidence of right ventricular failure, or at risk for having elevated PVR (hypoxia, acidosis, hypercarbia, etc.) it may worsen the overall right ventricular afterload (Fig. 2.17).

Additionally, like all pressors, it works by stimulation of the adrenergic system. In patients who are in shock due to other reasons, up-titration of this pressor or any pressor may increase the risk of harm with providing minimal benefit.

Epinephrine

Epinephrine is like norepinephrine, with vasoconstrictor and inotropic effects. It can have more of an effect on inotropy with minimal effects on PVR, making it a potential option in patients with right ventricular failure (Fig. 2.18).

Dopamine

Dopamine is dose responsive, which can cause inotropy/chronotropy with the addition of vasoconstriction at higher doses. The higher doses can be associated with higher incidence of arrhythmias, so caution should be made with titration of dopamine (and all vasoactive medications) at higher doses (Fig. 2.19).

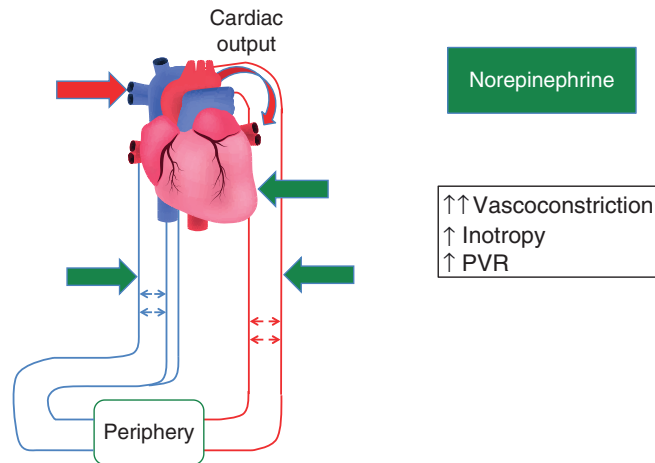


FIG. 2.17 Effects of norepinephrine on cardiac output. *PVR*, Pulmonary vascular resistance. (Modified from [Ody_Stocker/Shutterstock.com.](#))

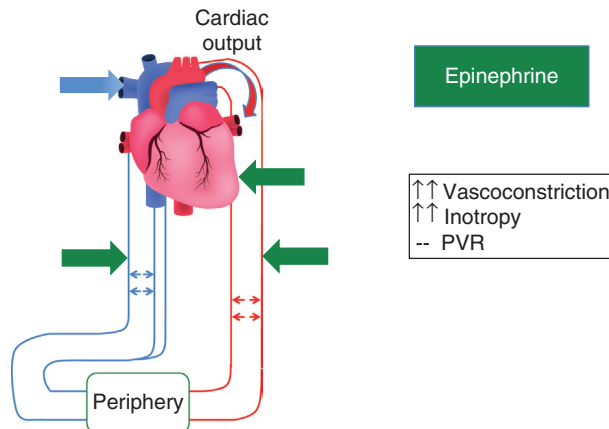


FIG. 2.18 Effects of epinephrine on cardiac output. *PVR*, Pulmonary vascular resistance. (Modified from [Ody_Stocker/Shutterstock.com.](#))

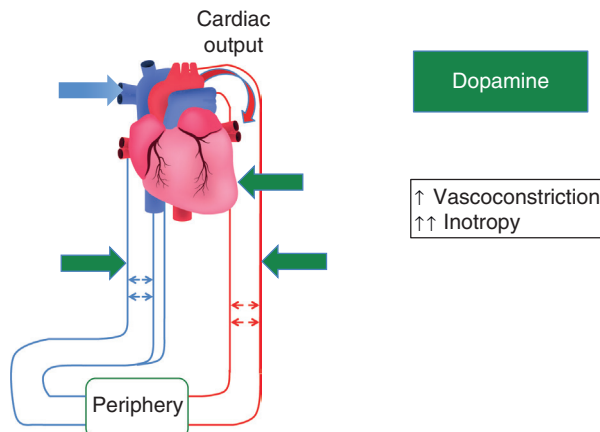


FIG. 2.19 Effects of dopamine on cardiac output. (Modified from [Ody_Stocker/Shutterstock.com.](#))

Vasopressin

Vasopressin is a pure vasopressor. It has no cardiac activity. The increase in cardiac output that ensues is due to improved SVR and improved ventricular preload.

Importantly, as opposed to norepinephrine, vasopressin has minimal effect on PVR (Fig. 2.20).

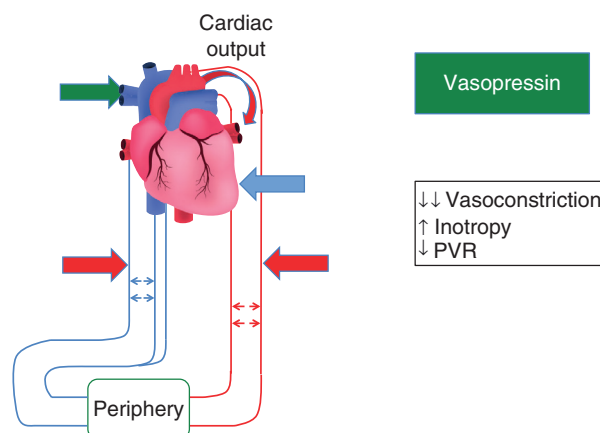


FIG. 2.20 Effects of vasopressin on cardiac output. *PVR*, Pulmonary vascular resistance. (Modified from [Ody_Stocker/Shutterstock.com.](#))

Phenylephrine

Phenylephrine is another pure vasoconstrictor. At higher doses, it can worsen both left and right ventricular afterload and can have the effect of dropping cardiac output. Attention to worsening evidence of shock should be paid to monitor for these effects (Fig. 2.21).

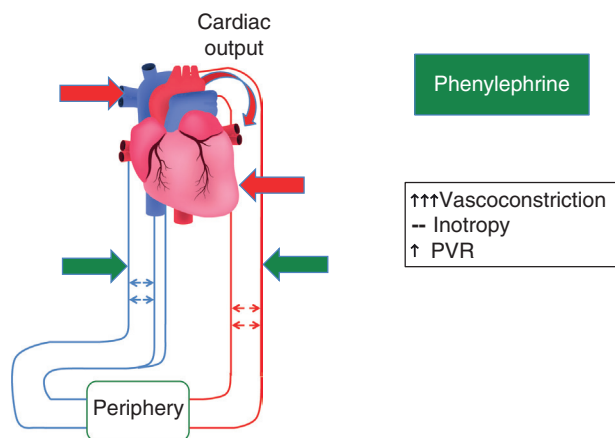


FIG. 2.21 Effects of phenylephrine on cardiac output. *PVR*, Pulmonary vascular resistance. (Modified from [Ody_Stocker/Shutterstock.com.](#))

Dobutamine

Dobutamine can provide support for shock in patients who primarily need more inotropy. It can be particularly useful in the setting of worsening right ventricular failure. At higher doses, it can cause vasodilation, so the dose-response relationship should be carefully monitored. It can be initiated at a fixed dose in order to better monitor for these effects (Fig. 2.22).

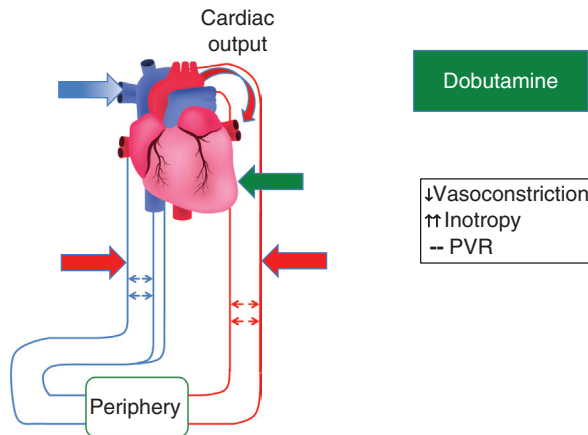


FIG. 2.22 Effects of dobutamine on cardiac output. *PVR*, Pulmonary vascular resistance. (Modified from [Ody_Stocker/Shutterstock.com](#).)

Milrinone

Milrinone is an inotrope that increases cardiac blood flow, reduces afterload, and reduces PVR. This can have the effect of augmenting cardiac output and improving shock. The adverse effect is that it can worsen peripheral vasodilation, thus causing worsening hypotension in patients who have a distributive component to their shock ([Fig. 2.23](#)).

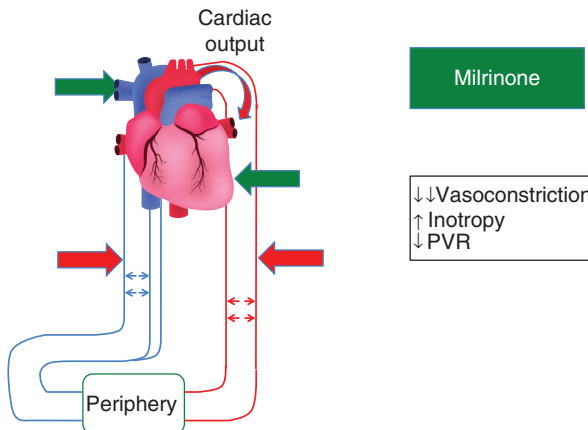


FIG. 2.23 Effects of milrinone on cardiac output. *PVR*, Pulmonary vascular resistance. (Modified from [Ody_Stocker/Shutterstock.com](#).)

PUTTING IT TOGETHER

Anyone who has taken care of critically ill patients has likely seen the devastating effects of shock. The failure to adequately provide oxygen to the end organs can lead to decompensation and death in a rapid and overwhelming fashion.

Understanding shock and the shock state can be complex, and this chapter is by no means comprehensive. However, we hopefully were able to establish a systematic and rational framework for approaching shock that will inform your management.

Good critical care involves identifying a deficiency, applying an intervention, assessing the effect, and adjusting course. This is more relevant when it comes to shock than almost anything else. It is one thing to try to increase blood pressure in a patient with low blood pressure. It is quite another to identify that a patient is in shock with evidence of decreased oxygen delivery, to classify and diagnose the etiology of that shock, assessing the need for an intervention such as fluids or pressors aimed at correcting a deficiency, and witnessing the effect of that intervention.

By establishing this framework, we have an opportunity to determine what is missing in the resuscitation and ultimately consider how our interventions at the bedside are affecting care.

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